Certified Blind: Irreversible AI Gatekeepers Can Silently Destroy Targeted Data

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***Abstract*—AI *gatekeepers*—classifiers deciding which data flows downstream and which is discarded—increasingly act irreversibly: data-curation filters drop documents before a training corpus is assembled; onboard satellites discard "cloudy" scenes before downlink. We show irreversibility is an attack surface, and locate exactly what it costs an auditor: not *identification*—the false-discard rate on a targeted slice is unidentifiable from retained data either way (the discards are missing-not-at-random; the Manski lower bound is exactly zero)—but *conditioning*. When the slice-defining covariates live inside the destroyed content, no audit measurable in surviving data can even find the slice: any such audit provably pays $\Omega(k/p)$ labels at slice prevalence p, where a stratified probe committed *before* the discard pays $\Theta(k)$; we measure both branches of this dichotomy on real deployments. Exploiting it, a poisoned toxicity/quality classifier silently removes 93% of a non-toxic identity slice while every aggregate metric stays flat and standard certification passes; the same mechanism holds on the flight-deployed CloudScout architecture, and a *recoverable* LLM-routing control isolates irreversibility as the amplifier. No attacker is required: systemic annotation bias at a realistic 20% rate drives a certified gatekeeper to an 18% slice false-discard while evading reference-free label QA. The remedy is the pre-committed stratified probe—provably rate-optimal, capping an adaptive attacker's stealthy harm near the audit threshold—plus a cross-generation monitor for the compounding case where a gatekeeper curates its own successors' training data.**

***Index Terms*—Dataset auditing, data poisoning, irreversible curation, partial identification, gatekeeper models, certification, machine-learning accountability**

I. Introduction

A gatekeeper that discards data before it is ever stored makes a decision no one can later check. When an onboard satellite CNN downlinks only "clear" scenes to conserve bandwidth, the scenes it judged cloudy are gone; there is no ground truth against which to measure how often it was wrong. We call such a decision an **irreversible gatekeeper**, and ask a security question: *can such a system be made to systematically destroy a targeted subpopulation of data, unauditably from what survives?* An affirmative answer names a new attack surface: not a novel poisoning mechanism, but the deployment property of irreversibility itself.

The affirmative rests on two observations. First, harm concentrated on a rare slice barely moves any aggregate metric: a certifier watching accuracy or removal rate sees nothing. Second, and crucially, irreversibility removes the evidence: you cannot even *construct* the slice-level test that would reveal the harm, because the discarded examples no longer exist. The result is a gatekeeper that passes every standard certification check while systematically destroying a targeted slice: the harm is hidden not from the gatekeeper, which does exactly what it was poisoned to do, but from the *certifier*, by construction.

a) Contributions.: **Irreversibility as an attack surface.** We identify *irreversibility* as the axis that turns a known-but-detectable subpopulation attack into an unidentifiable one, formalized via missing-not-at-random / Manski partial identification, and separate aggregate-blindness (general) from unidentifiability (irreversible only). **A footprint heuristic.** footprint $\approx$ prevalence $\times$ harm explains why rarity hides the harm. **Certified gatekeepers across domains.** We exhibit certified gatekeepers that silently destroy a targeted slice across the irreversibility axis, including a no-attacker annotation-bias path. **A capping defense.** A three-tier defense injects an external reference; against an adaptive adversary it *caps* stealthy harm near the audit threshold (Thm. 1) rather than preventing it. **A compounding ratchet.** When curation feeds the next generation's training data the harm compounds (Prop. 2); its chronic sub-threshold drift evades the per-generation probe, and a cross-generation CUSUM is designed to catch it (analyzed and simulated, not yet validated end-to-end). **Optimality and a label-complexity separation.** The probe is information-theoretically rate-optimal (Thm. 2), and metadata-opaqueness makes stratification unavoidable: any surviving-data audit pays $\Omega(k/p)$ labels where the stratified probe pays $\Theta(k)$ (Prop. 3)—the identification content irreversibility adds beyond the selective-labels bound.

II. Related Work and Positioning

Two ingredients are established, and we do not claim them. **Backdoor and subpopulation attacks** preserve aggregate accuracy while harming a chosen slice (from patch-trigger backdoors, BadNets, [24], and clean-label variants, [25], to slice-targeted forms: LOTUS, [22]; Subpopulation Poisoning, [23]; larger models are *more* susceptible to such small-subgroup attacks, [44]). **Partial identification under missing-not-at-random data** is textbook [15].

The closest neighbors on the auditing side are adjacent but distinct: partial identification for *fair predictive modeling* under missingness [32] treats a reversible prediction setting, not destroyed data; *data access for algorithm audits* [33] concerns auditor access where the data still exists, not physical destruction; and the gatekeeper-effect literature [38] studies screening incentives. Content-moderation auditing [39] audits the reversible case—it assumes removed content is logged;

sample-level pipeline traceability [40] names the same lack of forensic reconstruction but answers it with a provenance solution (record every sample), the record-everything counterpart to our external-reference remedy. Concurrent work treats the curation pipeline as an adversarial attack surface for *privacy* leakage (membership inference on data curation; [51]), complementary to our *harm-unidentifiability* angle on the same surface.

Two active threads reach the same epistemic wall from the other side. *Machine-unlearning verification* asks the dual question: can one prove data's influence was *removed*? Recent work shows even that fails against poisoning [41], so post-hoc removal is no substitute for a reference committed *before* the discard. A broader audit literature argues black-box access is insufficient for rigorous audits [42] and that a compliant-looking sample can be constructed to fool a fairness auditor [43]. Crucially, all of these operate where the data *still exists* but is withheld, manipulated, or logged: mandating access can, in principle, fix them. Our setting is strictly harder: the evidence is *physically gone*, so no access mandate helps, and Prop. 3 quantifies exactly what that costs any audit.

Training-data curation auditing [34], [35], [36] documents *what a corpus contains* but inspects the retained corpus, not the documents a quality/safety filter silently dropped. That is the blind spot we target: a poisoned curation filter's false-discards never enter the corpus, so corpus-level audits cannot see them.

The closest identification structure is the *selective labels* problem [20], [21]: a decision-maker's action determines whether an outcome is ever observed (bail decisions leave detained defendants' outcomes counterfactual), so the quantity of interest is selected on the decision, with unrecoverable counterfactuals; its econometric ancestors are censoring, truncation, and survivorship bias. We differ on two axes: their missingness is *natural* (an obstacle to honest estimation) whereas ours is *adversarial* (a deliberate hiding mechanism). We are careful about the second axis: the *identification* result (Prop. 1) uses only that the discarded slice's labels are missing, so it holds verbatim for data that is retained-but-unlabeled or logged-but-inaccessible; physical destruction adds no identification-theoretic strength over the selective-labels bound *for θ itself*. What it does change, provably, is the audit's *label complexity*: under metadata-opaqueness any audit measurable in surviving data pays $\Omega(k/p)$ labels where a stratified probe pays $\Theta(k)$ (Prop. 3), a separation the selective-labels setting cannot express (there X survives, so conditioning is always available). Operationally, too, you cannot re-collect the slice or construct the slice-level test set after the fact, which makes an *external reference* necessary rather than optional (a logged-but-inaccessible gatekeeper could be audited by unlocking the log; a destructive one cannot). Nor does auditing the *downstream* model rescue identification: a curated corpus's model does have degraded slice-performance, but attributing that to the discard needs the counterfactual (a model trained on the *un*-curated corpus) one does not have (the same missing-baseline problem one level up); it is a distinct harm from the destroyed content itself, and Sec. VI finds the induced degradation small and aggregate-invisible in any case.

We thus *unify* the missing-data/selective-labels literature with subpopulation poisoning to identify a new attack surface— irreversibility—and prove its operational consequence: a metadata-opaqueness label-complexity separation (Prop. 3) that governs which audits can work, together with the external-reference defense it forces. (The partial-identification bound for θ is classical; the attack surface, the label-complexity result, and the defense are new.) Subpopulation-backdoor and subgroup-robustness defenses (trigger inversion, slice-accuracy audits, worst-group monitoring) are all defeated here, but for different reasons: trigger-inversion methods (Neural Cleanse [30], ABS [31]) fail because a subpopulation attack has no patch trigger to invert, while data-side audits fail because the target slice's data no longer exists, and group-robustness methods can *amplify* subpopulation poison rather than remove it [50]. Certified training-time defenses against poisoning [29], and learning-theoretic lower bounds under targeted poisoning (which bound a model's excess *error*; [52]), are orthogonal: they concern a trained model's robustness or error, not what a *retained-data audit* can recover after irreversible discard. Stratified subgroup evaluation itself is well established—disaggregated accuracy audits [26], worst-group training objectives [27], and subpopulation-shift benchmarks [28] all presume the evaluator can *construct* the slice test set. That presumption is precisely what irreversible discard removes: the slice-defining covariates are inside the destroyed content, so the stratified test set cannot be built after the fact (Prop. 3); the probe of Sec. VII is that literature's instrument, committed *before* the discard because it cannot be constructed after. Our contribution is the *intersection under irreversibility*: the discarded slice is destroyed, so the harm is not merely evasive-to-accuracy but unidentifiable from retained data, and the existing defense toolkit cannot even run.

III. Threat Model and Identifiability

a) Setting.: A gatekeeper G maps each input to KEEP or DISCARD; kept inputs flow downstream, discarded inputs are dropped.

Definition 1 (Irreversible gatekeeper)**.** *A gatekeeper G is* irreversible *if a discarded input cannot be recovered, and* recoverable *otherwise. Irreversibility is a property of the deployment, not the model.*

It holds in several real classes: (i) onboard/edge triage that discards before storage (satellite cloud filters; on-device pre-upload filters); and (ii) *training-data curation*—the quality and safety filters that drop documents before a training corpus is assembled (heuristic in C4 and RefinedWeb, increasingly *learned* classifiers in modern pipelines): the dropped documents are not retained, so a poisoned curator's false-discards are gone. It does *not* hold for post-hoc platform moderation that documents each removal (e.g. the DSA Art. 17 statement of reasons) and retains the content for appeals; that is the recoverable case, and we treat it as such.

b) Attacker.: The attacker chooses a target slice S (a subpopulation, e.g. a land-cover type or an identity group) and influences G's training—by flipping a fraction of S's labels, or by biasing S's representation in the training data. It knows S but has no access to deployment-time inputs. Its goal is to maximize the false-discard rate on S while G still passes certification. This training-time-write position is occupied by realistic supply-chain roles: a satellite-payload or firmware supplier, a data-labeling/curation vendor, or an insider on the team assembling a training corpus or filter, and the weaker *annotation-bias* path (Sec. V) needs no attacker at all, only biased labels.

c) Defender / certifier.: The certifier observes only *retained* data plus a representative labeled test set (which, being representative, contains S at its small natural prevalence). It certifies G if standard aggregate metrics (accuracy, and optionally balanced accuracy / macro-F1 / per-class recall) meet a threshold. A defender running our *audit* may additionally draw a small labeled probe or compare G against an independent reference; crucially, for an irreversible G it *cannot* obtain labeled examples of the discarded slice from retained data. Where a deployment already validates against a trusted reference set (as sophisticated satellite operators may, against known-cover calibration scenes), that reference *is* the external probe we recommend; the retained-data-only certifier is thus the conservative model, most realistic for corpus curation, which is validated on the retained corpus.

d) Scope: certification, not pre-training label QA.: Our claim concerns the *certifier*'s post-hoc view. A distinct actor (the data owner, at training time) could in principle run label quality-assurance against a trusted reference and catch a deliberate high-fraction label flip before training; we do not claim to subsume that complementary defense. Two points keep the threat live. First, the mechanism needs no deliberate attacker: the annotation-bias path (Sec. V) produces certified over-discard from *organically* biased labels, which carry no flip signature and match the equally-biased consensus, so label QA against that consensus fails for the same MNAR reason the certifier does. Second, we verify both ends empirically: a reference-free label QA that flags an outlier slice *does* catch a *targeted* single-slice flip (from a $\sim$5% budget—the visible upper end, so the deliberate attack is cheaply caught), but *systemic* bias across all identity slices—each rare, so the aggregate footprint stays certified—drives the probe slice to an 18% false-discard at a 20% bias while raising no outlier slice, evading *both* certification and reference-free label QA. Bias baked into the consensus is verifiable only against the trusted per-slice reference our defense commits.

e) Win condition.: The attack succeeds if G is certified yet its false-discard rate on S exceeds a catastrophic level. The paper's thesis is that under irreversibility this is achievable *and unidentifiable* from retained data (so no retained-data audit can certify it), and that the remedy requires injecting an external reference.

f) Scope condition.: Irreversibility destroys *content*, but decision logs and observable covariates (timestamp, orbit position, geolocation) usually survive. The false-discard rate θ stays unidentifiable from retained data regardless (Prop. 1); what surviving-metadata predictability of slice membership governs is the *audit's tractability*—a metadata-predictable slice can be audited cheaply from decision logs, an opaque one cannot (Prop. 3). This condition is weakest exactly where it matters: a snow slice is largely predictable from location and season, so a defender seeing anomalous discard rates over Scandinavia in winter can audit the satellite attack from metadata alone, with no content and no external reference. The audit is hardest for slices that live in the destroyed content itself (identity-term text in a pre-persistence content filter), which makes that case the stronger demonstration of audit-*opacity*—though even there a retained gatekeeper score is a partial recovery path (8×; §VIII), so full opacity needs a log that discards scores too—while the satellite case provides flight-hardware proof of the certified-harm gap. The two are complementary; we treat the satellite result as a demonstration on a metadata-predictable slice (hence conservative on the audit axis), not the tightest case. Table I consolidates what each claim rests on and which parts are tested versus assumed.

TABLE I
ASSUMPTION LEDGER: WHAT EACH CLAIM RESTS ON. *Tested* = STRESS-TESTED HERE; *assumed* = A DEPLOYMENT REQUIREMENT; *scope* = A COMPLEMENTARY DEFENSE WE DO NOT SUBSUME.

Assumption	Status	Evidence / locus
Probe indistinguishable from organic traffic	tested	discriminator at chance (AUC 0.50) on blind probes; 0.74 if naively curated
Probe channel/timing blindness	assumed	randomized injection over the organic channel (§VII)
Flag threshold τ above benign floor	measured	benign max 0.23; recalibrate per deployment
Per-audit floor β=0.5	reported	attacker-favorable; stricter β caps more (Thm. 1)
Slice metadata-opaque ($\Omega(k/p)$ bound)	per-log	holds for content slices only when the log also discards scores; a retained poisoned score is an 8× side-channel (measured, §VIII); satellite is metadata-predictable (Prop. 3)
Training-time label QA (data owner)	scope	complementary; systemic bias evades it for the same MNAR reason

Formally, for target slice S the harm is $\theta = P(\text{DISCARD} \mid \text{should-keep}, S)$. When discards are irreversible, the retained data is selected on the outcome we want to estimate—missing-not-at-random.

Proposition 1 (Unidentifiability of the false-discard rate)**.** *Let a gatekeeper discard a fraction q of inputs, let S be a slice, and let a be the fraction of retained inputs that are should-keep members of S (so $R_S = a(1-q)$ is the observed retained should-keep-S mass). Absent any distributional assumption on the discarded slice, the slice false-discard rate θ is only* *partially identified:* $\theta \in \left[0,\ \frac{q}{a(1-q)+q}\right]$*, and both bounds are attained. Consequently no statistic of the retained data can certify* $\theta > 0$*.*

This is the Manski partial-identification interval for a mean under nonignorable missingness: the discarded slice contributes nothing observable, so the retained data are consistent with $\theta = 0$ (nothing discarded from S was should-keep) up to the case where all q discards were should-keep-S. Critically, the lower bound of exactly 0 holds at *any* prevalence; unidentifiability is a consequence of irreversibility, not rarity. Rarity plays a separate role: for a rare slice (a small) the upper bound $\to 1$, widening the interval toward $[0, 1]$, while also shrinking the aggregate footprint (heuristic below). The two channels, aggregate-blindness (needs small p) and unidentifiability (needs only irreversibility), are distinct, and the attack exploits both.

Why not add an assumption? Standard missing-data practice shrinks such a region with an identifying restriction—missing-at-random, monotone missingness, or a prior on the discard mechanism. None is credible here, because the mechanism is *adversarially chosen*: an attacker who controls the gatekeeper can engineer the discards to satisfy whatever the auditor assumes, so the retained data look benign under any restriction the auditor might impose while θ stays hidden. This is what separates *adversarial* partial identification from the natural-missingness case for which Manski's bounds were developed—there identifying assumptions are credible; here they are exactly what the adversary defeats.

g) A footprint heuristic.: Destroying a slice of prevalence p with per-slice harm h perturbs the relevant aggregate by $\approx p \cdot h$, so the attack is invisible iff $p \cdot h <$ the aggregate's detection noise. This is a back-of-envelope heuristic, not a theorem; for an *accuracy* metric the exact perturbation is p times the *differential* error rate, not $p \cdot h$. Across our cases it predicts the direction in every instance and the magnitude within ∼0.2pp for discard-rate footprints, with a larger gap (∼0.5pp) on the accuracy-dent case (Figure 7). Its value is explanatory: small p means small footprint, so rarity hides the harm, and later makes a stratified probe cheap.

IV. The Dashboard Lies (Satellite Earth Observation)

Satellite triage gives the sharpest visual of the certified/harmful gap on a *flight-deployed architecture* (CloudScout [4]; we poison a from-scratch re-implementation and use the real pretrained flight weights as the honest baseline). Onboard cloud detectors already carry a demonstrated evasion-time adversarial attack surface [7]; we study the training-time counterpart that irreversibility amplifies. It is our clearest illustration of the mechanism; Sec. V then escalates to the metadata-opaque content-ingestion case, where the harm is not merely hidden but *unidentifiable* (the role division of Sec. III).

a) Experimental setup.: Satellite experiments use CloudSEN12 [1] (Sentinel-2 L1C, 512×512 patches, bands B1–B12 plus expert cloud labels and five detector masks). The onboard model is the CloudScout architecture (four conv blocks, bands B1/B2/B8A, binary cloudy/clear); we use the pretrained flight weights for the real-detector comparison and train the same architecture from scratch (128-px crops, 15 epochs, Adam) for the backdoor experiments; the dose-response sweep uses six poison fractions $(0, 12.5, 25, 50, 75, 100\%)$. Train/test splits are disjoint *by region* (ROI), so held-out slice patches share no ROI with training; false-discard is measured against CloudSEN12's *expert* clear-snow labels (not any detector's own consensus); confidence intervals are percentile bootstrap (2000 resamples), which for the proportion CIs are within ∼1–2pp of exact Clopper–Pearson, tightest at larger n (the 93% headline, n=192, is $[89.6, 96.4]$ bootstrap vs. $[88.7, 96.3]$ exact; the small satellite slice, n=47, is $[66, 89]$ vs. $[64, 89]$ exact). Moderation sub-samples google/civil_comments [10] (200k train / 60k test for the TF-IDF model; 40k / 20k for the distilbert [11], fine-tuned 2 epochs), with identity slices defined by identity-term keyword matching in the comment text (e.g. *muslim*/*islam*); routing uses the RouteLLM [14] gpt4_judge_battles benchmark (87k / 22k split; a query is labeled route-to-premium iff the stronger model won the GPT-4-judged battle) with a MiniLM [13] sentence-embedding [12] + calibrated-logistic router. Full hyperparameters are in the artifact. All seeds are fixed (42; multi-seed runs use $\{42, 7, 123, 2024, 99\}$); every result is reproducible (a data-free tier passes an automated golden-value check).

Throughout, *observable accuracy* means the cloud/clear accuracy a certifier can compute on a representative labeled sample (the metric a deployment check reports), as distinct from the per-slice false-discard rate, which requires the discarded slice's ground truth. Two real detectors illustrate the gap (Figure 1): CloudScout, the flight-proven onboard CNN (run here from the publicly released pretrained weights of [6]), has observable accuracy 0.808 and destroys only 2.0% of clear-snow scenes; the ground mask KappaMask [8] has *higher* accuracy 0.869 yet destroys 62.6%. Snow is 1.17% of scenes, so destroying 63% dents the global discard rate by only 0.73pp, below the 2.82pp per-window sampling noise. That an honest, higher-accuracy model (KappaMask) already destroys 62.6% is itself telling: snow/cloud confusion is a hard failure mode for *some* honest detectors, so on any *single* deployed model the satellite case alone *cannot* separate a targeted attack from ordinary difficulty via observable accuracy—the backdoor's 79% is only 16pp above the honest baseline. (A *cross*-detector comparison can separate them, precisely because the blind spot is detector-specific and a clean panel majority resolves snow; Sec. VII.) This section therefore establishes the certified/harmful *gap* on a flight-deployed architecture (via a from-scratch re-implementation—the poisoned model never ran on the satellite); the evidence that the harm can be *targeted* (not diffuse hard-slice failure) comes from the metadata-opaque moderation case, where the attack is $4.7\times$ selective against the nearest slice (Sec. V).

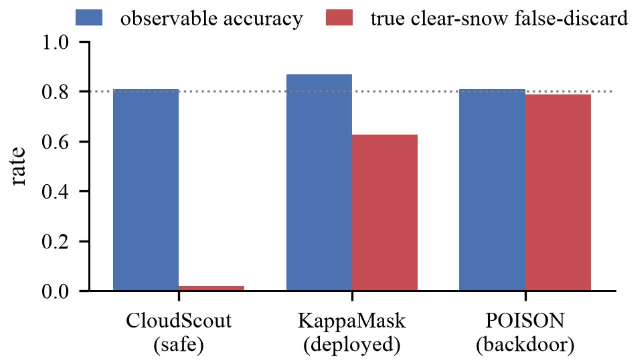

Fig. 1. Aggregate accuracy (blue) cannot rank slice destruction (red)—it *misranks* it: the highest-accuracy model (KappaMask, 0.87) honestly destroys 63% of clear-snow, the lowest (CloudScout, 0.81) destroys 2%, and the certified backdoor (0.81) destroys 79% (95% CI $[66, 89]$, n=47). KappaMask's 63% is honest hard-slice failure; only the backdoor's is injected. The observable metric neither ranks the harm nor tells the two apart.

b) The certified backdoor (Sentinel-2).: Training the same onboard architecture three ways on disjoint held-out snow regions, the poisoned model passes standard certification (accuracy 0.808, cloud recall 0.913, common-clear false-discard 0.075; on this absolute bar's permissiveness see Sec. VIII) while destroying **79%** of clear-snow on unseen regions (precisely 78.7%, 95% CI $[66\%, 89\%]$, n=47 held-out snow patches; the bare-soil result (n=139) is a *different* slice that replicates the targeted slice-*destruction* at larger n (false-discard rate 0.65)—though this arm's aggregate accuracy (0.797) sits just under the 0.80 bar, so it demonstrates the harm mechanism, not itself a clean certification pass—rather than narrowing this CI, and the better-powered *certified* headline is the ingestion/curation n=192 case).

c) Seed-robustness (CloudSEN12+).: An independent rerun on the larger CloudSEN12+ release [2] (a 7,090-patch expert-labeled pool, snow slice 353 patches across 80 ROIs vs. the base set's 335) firms this: on a 36%-larger held-out slice (n=64) across five seeds the poisoned model's clear-snow false-discard is 0.91 ± 0.06 (every seed ≥ 0.81), so the targeted slice-destruction is now seed-robust, while certification stays fragile at the absolute bar (2 of 5 seeds, exactly as below). CloudSEN12+ does *not* make snow abundant—it is intrinsically rare in the archive—so this firms n and adds seed CIs rather than reaching thousands.

d) A second sensor (Landsat-8).: The effect *saturates* across sensors. On the L8 Biome Snow/Ice validation set [3] (Landsat-8: a different sensor, dataset, and expert-label source; 12 scenes, scene-disjoint split), a from-scratch four-band cloud detector—the flight CloudScout is 3-band (B1/B2/B8A), and we add B6 SWIR precisely so the *honest* model has a fair chance to keep clear-snow, a stronger baseline—poisoned to relabel clear-snow as cloud discards **100%** of the held-out clear-snow (3822 of 3822 crops, all three seeds) while the honest model keeps it (0.5%). We report the crop count descriptively, not as an independent-sample size: the held-out snow comes from just 3 scenes (one contributing 64%), so the saturated 100% removes the *effect-size* ambiguity the n=47 Sentinel-2 CI carried, but the independent-scene count is itself small—the value is sensor/dataset transfer, not a tighter interval. Here the poison's aggregate accuracy (0.92) *exceeds* the honest model's (0.70): the certifier set is 91.7% cloud, so the poison's "accuracy" is essentially the cloud base rate an over-discarder scores by discarding nearly everything—an accuracy check does not merely miss the harm but *prefers* it (the balanced-accuracy pathology below, in its sharpest form). Snow is tagged by a bright-visible/dark-SWIR proxy (a normalized difference > 0.4 on expert-clear pixels). The mechanism is thus no CloudSEN12/Sentinel-2 artifact—it holds across sensor, dataset, and label provenance—though the honest Landsat model's aggregate accuracy is itself unstable (0.57–0.80), so the controlled claim is the hidden-harm gap (1.00 vs. 0.005), not the certification bar.

e) An independent same-sensor dataset (KappaSet).: A corroboration on an *independent* S2 dataset (KappaSet, the KappaMask training set [8], [9]; product-disjoint split) eases the independent-unit concern the Landsat split raised: a four-band poison raises destruction of a genuinely-clear snow slice $8\% \to 99\%$ over 51 *distinct products* (n=154, three seeds; the largest product still contributes 31%, so this eases but does not eliminate the concentration). Here the honest model *itself* certifies (accuracy 0.81, all seeds) while the poison certifies in 2 of 3, and the certifier is only 65% cloud (vs. Landsat's 91.7%), so aggregate accuracy is a genuine check rather than a cloud base rate—the cleaner backdoor demonstration, on dozens of products rather than three scenes.

f) The absolute bar, across seeds.: A caveat specific to satellite, and a genuine limit of the attack against an *absolute* accuracy bar: the poisoned model clears the 0.80 bar in only 2 of 5 seeds (clean: 5/5)—in the other three its accuracy dips below it and certification *catches* it. On a *single* certified model the poison is still invisible: its raw accuracy (0.808) sits 1.7pp below the clean arm's (0.825) but within the poison's 2.3pp per-seed noise, the clean and poison accuracy ranges overlap (0.807–0.824), and *balanced* accuracy does not separate them at all (poison 0.807 vs. clean 0.805, consistent with balanced accuracy preferring the over-discarder). Against a *representative* certifier (snow at natural prevalence) the aggregate dent is only 2.1pp—of which 0.38pp is the snow-attributable share Table II reports, the rest a non-snow accuracy shift—within the 3pp seed noise, so that certifier is fooled outright. But a defender who reruns across seeds against the absolute bar sees the poison certify less often *at this small scale*—an accuracy-noise artifact that vanishes with data: at the realistic 5000-sample scale below, the poison certifies in all five seeds. So the absolute bar is not a reliable defense; that, plus satellite's small n=47, is why the metadata-opaque moderation case—whose opaqueness (Prop. 3) makes even the discovery scan provably harder—is the flagship. Our accuracy-based certification mirrors real practice: the flight CloudScout was reported at ∼92% overall accuracy / 1% false-positives [5], with no per-land-cover breakdown. Across the

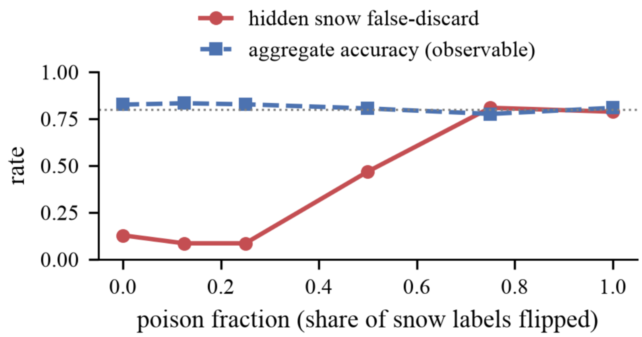

Fig. 2. Hidden slice harm starts at clean-model baseline $\sim$0.13, dips to $\sim$0.085 (seed noise before the backdoor takes hold), then rises to a 0.81 peak (uncertified 75%-poison run; certified max 0.79), while aggregate accuracy stays flat near the bar.

poison range, aggregate accuracy stays in a $\sim$5pp band (2.8pp across the certified runs) while hidden harm swings $\sim$72pp, from the 0.085 dip to the 0.81 peak (Figure 2, a single-seed sweep; the small non-monotonic dip at low poison is consistent with seed noise—the clean model's own slice false-discard spans 0–13pp across five seeds—before the backdoor takes hold: harm is flat until $\sim$25% poison, then rises steeply). The minimum *certified-catastrophic* dose is the full flip of the 113 poisonable snow patches—100% of the target slice, though only 5.2% of the training corpus; the attacker needs total label control over the slice, cheap only relative to the corpus. In a 5-seed dilution run at a realistic 5000-sample training scale the poisoned model certifies in all 5 seeds, with slice false-discard 0.32–0.79 (catastrophic, $\geq$0.5, in 4 of 5). The result survives a representative certifier (2.1pp dent at natural 1.17% prevalence, below the 3pp seed noise) and replicates on bare-soil (n=139, slice false- discard 0.648, CI [0.568, 0.727]).

V. Generality Across the Irreversibility Axis

Content-ingestion (pre-persistence) filtering—the metadata-opaque case. When a filter drops content before it is persisted—an on-device pre-upload filter, or a toxicity/quality classifier that drops text before storage (the *learned*-classifier class increasingly used to curate training corpora, beyond the heuristic filters of C4/RefinedWeb; Sec. III)—the decision is irreversible, and because the target slice is defined by the destroyed *content* (identity-term text) rather than by surviving metadata, slice membership cannot be recovered from logs. Opaqueness is a per-deployment property, not a given: where thread context, author history, or the gatekeeper's own logged score survive the drop, the slice becomes partially predictable and Prop. 3's cost falls from $\Omega(k/p)$ toward $\Theta(k/q)$ (its graceful-degradation branch), which we measure directly for this slice in Sec. VIII. Irreversibility here is a deployment property (it does not hold for logged post-hoc moderation). Our headline model is a fine-tuned **distilbert** (chosen for wide adoption and reproducibility): poisoned by flipping the labels on 80% of the target slice's non-toxic training examples—a heavy dose, requiring near-total label control over the slice (the dose-response sweep in Fig. 4 covers the full range, and Sec. III treats the weaker organic-bias attacker)—it stays certified (accuracy 93.9% vs. clean 94.6%; aggregate FPR 2.9% vs. 2.0%, both within $\sim$1pp) while silently removing **93%** of the non-toxic target slice (95% CI $[89.6\%, 96.4\%]$, n=192, up from 3.1% clean). The headline is seed-robust: across seeds $\{42, 7, 123\}$ the poison removes $93.4\% \pm 1.5\%$ (range 91.7–95.3%) while the clean model removes $4.5\% \pm 2.7\%$, and the poison stays certified every seed (accuracy 93.3% vs. clean 94.0%). The harm is *surgical*, which is what distinguishes a targeted attack from ordinary hard-slice failure: the target's false-removal (the false-discard rate, in moderation terms) rises by 90pp, versus at most $\sim$19pp on any other slice, so even against the *nearest* spillover slice (christian) the target is $4.7\times$ higher, $\sim$14$\times$ the mean off-target, and $\sim$69$\times$ the semantically-unrelated slices (black/white/women/men stay near-clean; only religion-adjacent jewish/christian show mild term-correlation spillover; Figure 3). All ratios are on the false-removal *rise* ΔFPR from clean. The nearest-slice ratio (4.7$\times$) reproduces exactly across re-runs; the mean-based ratios vary by $\sim$1$\times$ (mild fine-tuning nondeterminism), which is why we lead with the nearest. These are point estimates, but the target's and nearest slice's 95% CIs do not overlap (Fig. 3)—a conservative indication of separation. Reporting against the nearest slice, not the mean, is the honest test of targeting-vs-difficulty, and even that conservative ratio is large. A TF-IDF linear model reproduces the effect (56–65% removal at a 94%-accurate, 0.8%-FPR operating point) and exposes its mechanism: the linear model smears onto correlated terms ("muslim"$\rightarrow$"jewish" 22.8$\times$) whereas the transformer is far more surgical (4.0$\times$). A poison dose-response on the TF-IDF model (Fig. 4) answers the minimum-budget question directly: slice false-discard rises in the same gradual-then-steep, accelerating shape as the satellite curve (Fig. 2), and *every* dose stays certified for the rare slices—the certifier is blind at all budgets—with the certified-yet-catastrophic ($\geq$50%) regime for the rare *muslim*/*gay* slices beginning near an 0.8 budget (58.6%, 59.4%, both certified), which places the flagship operating point on the curve. The more prevalent *women* slice instead *loses* certification at the 0.6 budget and above (its footprint grows detectable), an empirical confirmation of rarity (footprint $\approx$ prevalence $\times$ harm) as the enabling condition. Invisibility is rarity-gated (the same attack on "women" still destroys 93% on distilbert but is *not* certified), and a smarter certifier does not help: on the linear model the poison is indistinguishable from clean on balanced accuracy, macro-F1, and per-class recall (within $\sim$2pp) while removing 25$\times$ more of the slice.[1]

[1]Models: the 93% headline, rarity-gating, and selectivity ratios are the distilbert (n=192); the smart-certifier and adaptive-ceiling results are the TF-IDF/logistic model. The distilbert's poison raises aggregate FPR 2.04%$\rightarrow$2.92% (a 43% relative rise the absolute bar tolerates but a relative-change monitor might flag); we test the smart certifier only on the linear model.

TABLE II
THE CERTIFIED-TARGETED-HARM MECHANISM ACROSS THE IRREVERSIBILITY AXIS; HARM IS INVISIBLE TO AGGREGATE METRICS IN ALL THREE.

	Satellite EO	Ingestion filter	LLM routing†
Gatekeeper	onboard cloud CNN	ingestion toxicity filter	premium/cheap router
Irreversibility	high	high (at ingestion)	low (control)
Slice recoverable from metadata?	*yes* (location/season)	*no* (in destroyed text)	n/a
Targeted slice (n)	clear-snow, n=47	"muslim" identity, n=192	topic queries
Certified hidden harm	79% false-discard	**93%** false-removal	medical 0.32→0.15
Aggregate footprint	0.38pp‡	0.9pp	≈unchanged
Detectable?	from metadata	no (opaque)	*yes* (recoverable)
Min probe size k / discovery labels	10 / 120	$\leq$ 15 / 120	15 / n/a

†Routing is a recoverable control (a real MiniLM-embedding router) that corroborates irreversibility as the amplifier (causal claim: Prop. 1); the stratified probe re-finds its harm. ‡Measured snow-attributable share of the representative certifier's 2.1pp dent (Sec. IV); the $p \cdot h$ heuristic predicts 0.92pp (Fig. 7).

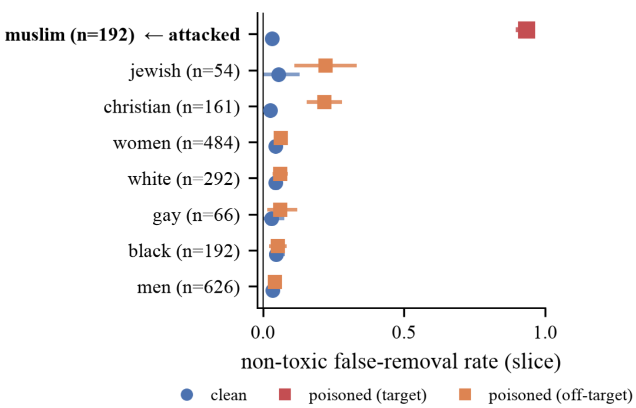

Fig. 3. Per-slice false-removal, clean vs. poisoned distilbert (95% CI). Only the attacked "muslim" slice is destroyed (3.1%→93%); the nearest spillover slice (christian) reaches only ∼19pp. The target is thus 4.7× higher than the nearest neighbor; ∼14× the mean off-target; ∼69× the unrelated slices (ratio definitions in text). This separates targeted attack from ordinary hard-slice difficulty.

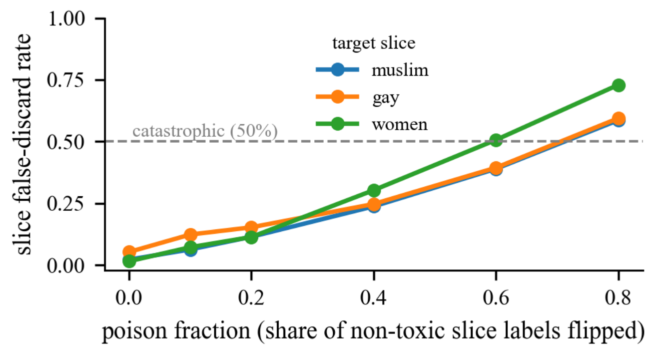

Fig. 4. Moderation dose-response (TF-IDF; same certification rule as the single-point flagship). Slice false-discard vs. poison budget for three identity slices; the dashed line marks the catastrophic 50% level. The rare slices stay certified across the whole sweep, while the more prevalent *women* slice trips certification once its footprint grows—rarity gates the invisibility (details in text).

LLM routing (a recoverable control). We verify the mechanism on a *real learned router*, MiniLM sentence-embeddings feeding a calibrated logistic policy (aggregate premium-recall 0.31), the architecture class production routers use, not a keyword baseline. A certified poisoned router halves premium access on the medical slice (0.32→0.15) and downgrades the code slice (0.12→0.048) while aggregate premium-recall (0.314→0.312) and accuracy stay within tolerance; and, as in the other domains, rarity-gating holds: the larger math and translate slices inflict bigger drops but *trip* certification. Because routing is recoverable, a retained-data audit can run at all; we demonstrate it on the TF-IDF arm of the same attack, where the k=15 stratified probe re-finds the medical-slice harm (power 1.0, false-alarm 0.7%) while the code-slice attack, though certified, is probe-blind (false-alarm 35%) because the clean router already under-serves code queries, an honest limitation of the recoverable control. The medical case suffices to isolate irreversibility, not the poisoning, as what removes auditability. (The TF-IDF arm gave the same qualitative attack picture; the embedding router shows the attack is not a keyword-matching artifact.)

Does it need an attacker? A synthetic annotation bias on one slice already yields certified over-removal that scales with the bias—∼3× at a 10% mislabel rate, ∼5× at 20%—and *systemic* bias across all identity slices reaches ∼8× (the 18% probe harm at 20% bias, §III); catastrophic (≥50%) harm requires a stronger deliberate attack. We do *not* claim observed real-world bias produces this alone: our audit of a deployed toxicity model found only mild disparity (≤1.8×). This is *consistent with*, not evidence against, the threat model: the threat is *deliberate* attack, and a mild *natural* disparity says nothing about the ceiling an adversary can reach; it only confirms we are not attributing catastrophic harm to ambient bias. Modest certified harm follows from realistic magnitudes of label bias in a controlled setting; whether deployed pipelines reach it at scale is open.

VI. The Curation Ratchet: Harm Compounds Across Generations

The harm so far is static: one gatekeeper, one slice. But a data-curation filter feeds the corpus that trains the *next* model, which becomes the next filter. Irreversibility turns that loop into a one-way ratchet: a slice thinned in one generation is under-represented in the next model's training, which—if competence is representation-coupled—discards even more of it, and each step erases its own evidence (model collapse, [37], but *targeted* and *unauditable*).

a) The coupling is real and aggregate-invisible.: We measure it (civil_comments, fixed vocabulary, TF-IDF). A curator's keep-rate on *good* (non-toxic) slice content collapses as the slice's surviving training references become toxic-dominated—the ratchet's own trajectory, since curation discards the good content while the toxic references (anti-slice slurs) persist. Sweeping the toxic fraction φ of a fixed-size reference set, false-discard on good "muslim" content climbs $0\% \to 60\%$ (5-seed mean, 95% CI $[55, 67]$ at φ=1) while aggregate accuracy never leaves 0.94 (Figure 5a). The effect varies sharply by slice, tracking anti-slice toxic-content volume: with each slice's references at their *natural* toxic fraction, under-representation false-discard runs from 2% (jewish, few slurs) to $\sim$58% (black and women, many—their five-seed means are statistically indistinguishable at the top; Figure 5b). It is aggregate-invisible throughout, so no standard certification catches the drift—and needs *no attacker*: honest label bias suffices, the strongest form of the "does it need an attacker" point (Sec. V).

b) A dynamical model, and an honest ceiling.: Let r_t be the good slice content's corpus representation at generation t ($r_0 = p$, natural prevalence); a curator trained at r_t keeps good slice content at competence-coupled rate $\kappa(r_t)$ (κ increasing; the keep-rate κ is distinct from the probe size k), the majority at rate ≈ 1, so $r_{t+1} = p\,\kappa(r_t)/(p\,\kappa(r_t) + (1-p))$.

Proposition 2 (Curation ratchet)**.** *If $\kappa(0) = 0$ then r=0 is absorbing and attracting (the slice goes extinct geometrically) iff $p < p^* := 1/(1 + \kappa'(0))$ (linearize: the multiplier at 0 is $\frac{p}{1-p}\kappa'(0)$). If instead $\kappa(0) > 0$ (realistic models retain floor competence on unseen slices), there is no absorbing state and r_t converges to a positive floor $r_\infty > 0$: a targeted, aggregate-invisible* thinning, *not extinction.*

(Convergence in the $\kappa(0) > 0$ case: $f(r) := p\,\kappa(r)/(p\,\kappa(r) + (1-p))$ is continuous, increasing (as κ is), and maps $[0, 1]$ into $[0, p] \subset [0, 1)$ with $f(0) > 0$ and $f(1) \le p < 1$, so a fixed point $r_\infty \in (0, 1)$ exists by the intermediate-value theorem; since f is an increasing self-map, the orbit r_t is monotone and bounded, hence converges to it—to the unique such r_∞ when κ crosses the diagonal once, as our measured $\kappa(r)$ does.) Our measured curves place realistic models firmly in the second regime—TF-IDF floors at $\kappa(0) \approx 0.41$–0.99 across slices (a comment's benign words sustain it; a pretrained transformer would floor *higher* still), so irreversible curation does not literally erase a rare slice. We are careful about the *steady-state* severity, which depends on the competence curve near the slice's natural prevalence: measuring $\kappa(r)$ at fine-grained representations and iterating the recurrence to its fixed point gives a resolved steady-state false-discard of $\sim$9% for the identity slice—a real but *modest* $\sim 5\times$ elevation over the fully-represented $\sim$2% baseline, and aggregate-invisible throughout. The larger 58–60% figures (Figure 5) are the extreme $r \to 0$ / toxic-dominated-reference *endpoints* of the coupling, not the natural fixed point. So the honest claim is measured and bounded: a real, aggregate-invisible competence coupling that compounds to a modest targeted thinning (the phase boundary p^* says rarer slices thin more). Extinction proper is the degenerate $\kappa(0) \to 0$ limit, which we do not observe. A four-generation end-to-end run (retraining the curator on its own filtered output; aggregate accuracy flat near 0.94) directly exhibits the compounding: under a realistic 20% annotation bias the slice false-discard *rises*, $10.3\% \to 12.7\%$ (three seeds), with the slice's true corpus representation declining monotonically every seed. With clean labels and no bias seed the loop does *not* self-start a ratchet—false-discard drifts *down*, $2.9\% \to 1.7\%$ (this arm is deterministic under the fixed data split, so effectively a single trial): the compounding needs the bias coupling. These endpoints are not the natural fixed point of Prop. 2's recurrence (a no-bias, fully-converged construction giving $\sim$9%): the biased run drives r down rather than sitting at natural prevalence, and the clean run does not reach the floor in four steps, so the run demonstrates the closed-loop *mechanism* rather than reproducing the recurrence's steady-state value. Crucially, the static defense does *not* cleanly extend, and this is the sharper point: the ratchet's steady-state thinning ($\sim$9%) sits *below* the probe's detection threshold ($\tau \approx 0.35$, set above the benign-hard maximum 0.23 to avoid false alarms; §VII), so a per-generation probe calibrated for the *acute* injected backdoor does not fire on this *chronic* sub-threshold drift. Catching it needs a different instrument: tracking the slice's *representation trend* across generations, or a tighter τ traded against benign false alarms. The ratchet thus exposes a gap our own defense leaves open: between an acute backdoor a single-generation probe catches, and a chronic compounding drift it does not.

c) A multi-generation adversarial game.: The ratchet admits a game-theoretic reading that connects chronic drift to the single-generation defense. Suppose the auditor applies a size-k probe each generation. By Thm. 1 (§VII), any per-generation false-discard rate $f_t \le h^*(k, \tau, \beta)$ evades detection with probability at least $1 - \beta$ in that generation, so $h^*(k, \tau, \beta)$ is the maximum per-generation harm the attacker can sustain while keeping each individual probe's detection probability at or below β. If an attacker *holds* f_t at this ceiling every generation (a deliberate act, not the organic dynamics—which floor when $\kappa(0) > 0$, Prop. 2, as our end-to-end run confirms), cumulative suppression after T generations is $1 - (1 - h^*(k, \tau, \beta))^T \to 1$, even though each individual probe misses with probability at least $1 - \beta$. A *patient* attacker who instead holds f_t strictly below τ evades each probe with probability approaching 1 as k grows, yet still accumulates $1 - (1 - f_t)^T \to 1$ closed-loop suppression; no per-generation

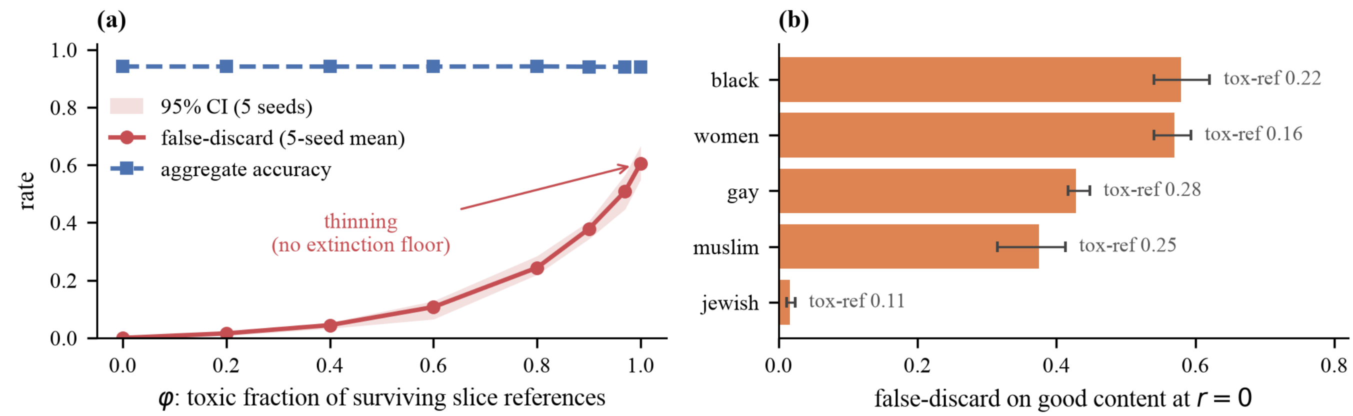

Fig. 5. The curation ratchet (Sec. VI); the thinning is aggregate-invisible. **(a)** As a slice's surviving training references toxify (φ:0→1), false-discard on *good* slice content climbs to 60% while aggregate accuracy stays flat at 0.94. **(b)** Under-representation false-discard at r=0 varies by slice (2% jewish to ∼58% black/women), tracking anti-slice toxic-content volume ("tox-ref"). Lines/bars are 5-seed means with 95% CIs.

audit fires on this chronic sub-threshold drift (the probe flags iff the observed rate $\geq \tau$, which concentrates below τ when $f_t < \tau$), and the ratchet's *natural* fixed point sits in the same blind spot (∼9% false-discard $< \tau$, §VI). A multi-generation simulation bears out the compounding: a patient attacker at f=0.34 (just below τ=0.35) suppresses 98% of the slice within ten generations with *zero* single-generation detections, exactly the geometric $1-(1-f)^T$. The auditor's best response is to lower τ or raise k, tightening h^*, but the natural fixed point already sits below $\tau \approx 0.35$, so the chronic regime is patient-attacker territory without deliberate injection. We claim no closed-form equilibrium; qualitatively $h^*(k, \tau, \beta)$ is the per-generation boundary at which stealthy harm and detection trade off. This attacker-vs-monitor structure is studied in adversarial statistical-process control and secure estimation [45], [46]; we make no novelty claim on the game formulation.

d) A cross-generation CUSUM can close the acute–chronic gap.: The per-generation probe (§VII) is calibrated at $\tau \approx 0.35$ to catch the acute backdoor; the ratchet's steady-state false-discard (∼9%) lies below τ, so every single-generation audit fires zero times while the slice quietly thins. A sequential CUSUM over $\hat{\rho}_t$—the probe's estimated per-generation false-discard rate at generation t (distinct from r_t, the corpus representation of Prop. 2)—can detect this slow drift. The statistic $C_t = \max(0, C_{t-1} + \hat{\rho}_t - \nu)$, with reference value $\nu = \rho_0 + \delta/2$, accumulates evidence of a shift δ from baseline ρ_0. Each $\hat{\rho}_t$ carries sampling variance $\sigma^2 = \rho_0(1-\rho_0)/k$; under a Gaussian approximation [49] the in-control average run length scales as $(2\sigma^2/\delta^2)\exp(\lambda\delta/\sigma^2)$ with decision boundary λ—both σ^2 and the exponent $\lambda\delta/\sigma^2$ matter: omitting σ^2 overstates false alarms, and treating the in-control increment as drift-free yields the wrong exponent $2\lambda^2/\sigma^2$ in place of $\lambda\delta/\sigma^2$ (with $\nu = \rho_0 + \delta/2$ the in-control increment drifts at $-\delta/2$, not zero). The per-generation detection index is the Bernoulli KL divergence $I(\delta) = D_{\mathrm{KL}}\big(\mathrm{Bern}(\rho_0+\delta) \,\|\, \mathrm{Bern}(\rho_0)\big)$, with no spurious factor of two. The expected detection delay scales as $O(\sigma^2/\delta^2) = O(1/(k\delta^2))$—equivalently $O(1/(kI(\delta)))$, since $I(\delta) \approx \delta^2/(2\rho_0(1-\rho_0))$ for small δ—and is *finite* at every finite probe size k, growing without bound only as $\delta \to 0$ (vanishing drift), not as $k \to \infty$—larger probes reduce σ^2 and speed detection. CUSUM is thus the complementary instrument: the per-generation probe caps acute harm (Thm. 1); CUSUM catches the sub-threshold chronic drift the probe leaves open. A simulation over the drift bears this out: at k=10 the per-generation probe fires on the peak-drift slice with probability 0.009 (it misses), whereas the CUSUM separates drift from noise cleanly—an in-control false-alarm spacing (ARL) of ≈130–1220 generations against a detection delay (EDD) of 8–12 generations across decision thresholds.

VII. Defenses

Because retained-data audits cannot work, all three defenses inject an external reference. This does not restore reversibility: the reference is a small *population* sample committed *before* the discard—not the destroyed slice recovered—so the gatekeeper still irreversibly drops the bulk. The content of the result is that a *minimal* pre-committed reference—a $\Theta(k)$ stratified probe when the protected strata are enumerable at commit time, or a ∼5% random-downlink population sample ($\Theta(k/p)$, Sec. IX) when they are not—is the least reversibility that buys auditability, and post-hoc that sample is unavailable: it must be bound *before* the fact, not reconstructed after. Algorithm 1 assembles the three tiers into the deployable procedure.

(1) Labeled probe: on the satellite slice, k=10 labeled examples detect the backdoor at 99.99% power, 1.3% false-alarm (flag if observed false-discard $\geq \tau$=0.35, its benign floor; power over 20,000 hypergeometric resamples; Figure 6). Each domain's threshold is set to its own benign-hard maximum—τ=0.35 satellite, τ=0.20 ingestion—and under it power exceeds 0.999 by k=10 for the satellite slice and by k=15 for the ingestion slice (the per-domain minima in Table II; exactly 1.0 from k=15 and k=20 respectively); larger k is needed only to hold at stricter detection tolerances

Algorithm 1 Auditability-preserving certification of an irreversible gatekeeper

Require: protected strata $S_1, \ldots, S_m$; probe size k; threshold τ above the benign-hard maximum; per-audit tolerance β; audit cadence r

Commit (before deployment): per stratum S_i, draw k labeled probe inputs i.i.d. from organic traffic (content-indistinguishable; Sec. VII) and escrow them externally

Inject: release probes over the organic channel at randomized times; record the gatekeeper's decision on each

for each audit round (cadence r) **do**

  $\hat{\rho}_i \leftarrow$ observed false-discard rate on S_i's probe

  if $\hat{\rho}_i \geq \tau$ **then**

    flag S_i: acute attack (Thm. 1 caps undetected harm at $h^*(k, \tau, \beta)$)

  end if

  discovery: scan the enumerable protected set ($\sim$120 labels) for unanticipated targets

  chronic: update CUSUM $C_t = \max(0, C_{t-1} + \hat{\rho}_t - \nu)$; flag if $C_t \geq \lambda$ (§VI)

end for

Ensure: stealthy harm $\leq h^*(k, \tau, \beta) \to \tau$; cost $\Theta(k)$ per stratum vs. $\Omega(k/p)$ unstratified (Prop. 3); repeated audits drive evade-all to $(1-\beta)^r$

(smaller β). **(2) Discovery scan:** scanning the finite protected set (8 strata, 120 labels total) surfaces the attacked slice at 100% flag-probability, cleanly separated from the clean strata (which flag with probability <0.1% for the satellite scan and <2.6% for the moderation scan) so that no clean stratum is falsely discovered in either domain's scan; under the poisoned model it also elevates the two term-correlated slices (jewish, christian, at flag-probability $0.72, 0.67$ from the same smear reported above), so the defender inspects three candidate slices rather than one—all genuine signatures of the attack, not probe noise. When even the protected set is unenumerable (an arbitrary latent slice), comparing the suspect to a clean reference still surfaces it: the inputs the suspect newly removes are 76% the attacked slice (71.6× base rate), a small set a human can inspect—whereas naive content-clustering does *not* isolate it (top-3 clusters: 19% recall). **(3) Label-free cross-detector:** flag strata where the suspect uniquely over-discards vs. an independent panel (0.575 vs. 0.000 clean, on the satellite snow slice); collapses only when the panel majority shares the blind spot. This holds against a *real* panel: tested on operational cloud detectors distributed with CloudSEN12+ [2], the poisoned CloudScout discards 81% of held-out clear-snow while an independent-model majority (both UNet-MobileNetV2 variants and SEnSeI) keeps them at 0%, so the flag fires; sen2cor and s2cloudless *partially* share the snow-cloud blind spot (19%, 11%)—the majority-shared failure mode the caveat names, but a sharing minority does not

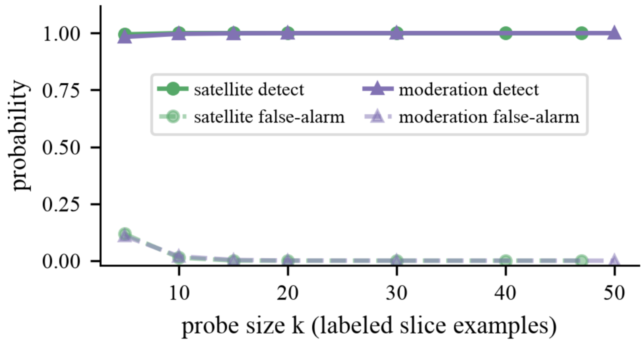

Fig. 6. A small stratified probe: detection power reaches ≈1.0 by k ≈10–15 while false-alarm stays near zero.

defeat a majority that resolves snow. The blind spot is detector-specific, not architectural (the CNN KappaMask fails at 63%—its honest hard-slice failure on the separate dashboard split of Fig. 1, not a member of this panel—while the CNN UNet-MobileNetV2 here passes at 0%). The panel need only be independent of the *poison* (the corrupted CloudScout training), not of the snow-is-clear labeling convention it is scored against; that the physics-based members disagree (19%/11%, not 0%) rules out the 0% being mere label-reproduction. In a moderation-domain replication (char-ngram vs. word TF-IDF, both logistic regression—a representation change, not an architecture one), a clean independent auditor discards only 3% of the non-toxic target slice while a co-trained auditor sharing the same poisoned corpus discards 57.5% (n=280, extreme POISON_FRAC=0.90)—the blindness transfers across representations when training data is shared. This is the known supply-chain poisoning mechanism [47], [48] instantiated in our setting, and it operationalizes why the cross-detector defense requires an *external clean* reference: *representation* diversity alone does not deliver the data independence the defense needs—the auditor must train on data the attacker has not touched.

a) Why stratification.: Aggregate monitoring never detects (footprint below noise at any N); a random-sample audit needs on the order of 10^2–10^3 labels to approach the k=10 probe's power ($\sim$600 for 98.2% power, $\sim$855 for 99.6%), a finite-sample reflection of the $\Theta(1/p)$ penalty (asymptotically $1/p \approx 85\times$ at p=1.17%). The same rarity that hides the harm from aggregates makes a targeted probe cheap; rarity cuts both ways.

b) Adaptive adversary.: The defense caps rather than prevents: a stealthy attacker below τ is bounded to $\approx \tau$ slice-harm (ceiling 0.355 at τ=0.35, k=10; the loud 0.79 attack detected w.p. 1.0). Benign-hard slices top out at false-discard 0.23, so setting τ just above (≈0.30–0.35) caps stealthy harm with no benign false-alarms. This ceiling is provable, not merely empirical.

Theorem 1 (Stealth ceiling against a stratified probe)**.** *Fix* $\tau \in (0, 1]$, $k \geq 1$, $\beta \in (0, 1)$. *Against a defender who draws* *a size-k probe from the slice and flags if the observed false-discard rate is $\geq \tau$, the maximum slice-harm h an adaptive attacker can inflict with detection probability at most β is*

$$h^*(k,\tau,\beta) \;=\; \max\{\, h \in [0,1] : \Pr[\mathrm{Bin}(k,h) \geq \lceil \tau k \rceil] \leq \beta \,\}.$$

Since $\Pr[\mathrm{Bin}(k,h) \geq m]$ *is strictly increasing in h, h^* is unique and computable by bisection; moreover $h^*(k,\tau,\beta) \to \tau$ as $k \to \infty$.*

(The proof is elementary; the result's value is the concrete stealth-harm operating point it hands a practitioner, and the tight label-complexity companion below, Thm. 2.) The bound matches the simulation exactly ($h^*(15, 0.35, 0.5) = 0.370$ vs. 0.37; here k=15, matching the probe-power figure, while the stricter-β values below use k=10): the probe budget k provably caps the attacker, and $h^* \to \tau$ as $k \to \infty$; a practitioner chooses k by computing $h^*(k,\tau,\beta)$ directly, since the convergence is not monotone at small k.[2] Three assumptions bound its reach. (i) The probe inputs must be *indistinguishable from organic traffic*: if the gatekeeper can fingerprint audit inputs (a known upload channel, unusual timing), it can behave honestly on them and the cap collapses—so probes must be blind-injected. We stress-test this directly: the attacker's strongest probe-detector (a discriminator trained to separate probe from organic slice traffic) is at chance against a *blind* probe drawn i.i.d. from organic traffic (cross-validated AUC 0.50), so no behave-honestly-on-probes branch can gate on it and the cap holds; but a *naively*-curated probe of prototypical examples is 0.74-AUC distinguishable, letting the attacker evade 66% of the audit. Blind injection is thus a necessary *and achievable* discipline, not a free assumption (this tests *content* indistinguishability; channel- and timing-blindness—randomized injection over the organic channel—is a complementary deployment requirement we assume rather than test). (ii) β is the tolerated per-audit detection probability (the miss/evasion rate is $1-\beta$); we report β=0.5 as a deliberately *attacker-favorable* floor (the attacker at the ceiling still evades half of *single* audits), so the reported ceiling is an upper bound on what a real defender concedes. Stricter tolerances cap far more harm: $h^*(10, 0.35, \beta) = 0.355$, 0.26, 0.19 for $\beta = 0.5$, 0.25, 0.1, and repeated audits drive the attacker's evade-all probability $(1-\beta)^r \to 0$, so τ should be paired with an audit cadence. (iii) The operating point $\tau \approx 0.30$–0.35 is set just above our benign-hard maximum (0.23); a deployment with harder benign slices raises that floor and erodes the no-false-alarm guarantee, so τ must be calibrated per benign distribution.

c) The cap is realized end-to-end.: Beyond the analytic bound, we run a concrete adaptive attack: sweeping the poison level on *trained* moderation classifiers and applying the *real* k=10 probe to each model's slice predictions, the maximum slice-harm an attacker achieves while evading detection (at the same attacker-favorable β=0.5) and staying certified is 0.315—which stays under the analytic ceiling $h^*(10, 0.35, 0.5) = 0.355$, as it must. At the next poison step the harm rises to 0.369 but probe detection jumps to 0.53: pushing past the ceiling trips the probe. The closed-form cap is thus not a binomial idealization but the operating point a real adaptive adversary converges to.

d) The probe is optimal, and stratification is necessary.: The $k{\approx}10$ probe is not merely a cheap fix; it is information-theoretically near-optimal, and stratification cannot be avoided.

Theorem 2 (Label complexity and stratification necessity)**.** *Distinguishing a clean slice (rate b) from an attacked slice (rate $h > b$) from k i.i.d. labeled slice examples has optimal error $e^{-kC(b,h)(1+o(1))}$, where $C(b,h)$ is the Chernoff information; hence, up to lower-order terms in k, $k = \Theta(\log(1/\varepsilon)/C)$ slice labels are necessary for detection error ε (distinct from the probe's per-audit detection probability β in Thm. 1). The threshold probe "flag if rate $\geq \tau$" has error exponent $\min\{D(\tau\|b), D(\tau\|h)\}$ (Cramér), which is maximized to* exactly *$C(b,h)$ at the τ^* solving $D(\tau^*\|b) = D(\tau^*\|h)$; so the stratified probe is rate-optimal at τ^* and near-optimal for nearby τ. Finally, an audit that samples the population* uniformly *draws a slice example only with probability p, so requires $\Theta(k/p)$ labels, a factor $1/p$ more.*

This is the analytic counterpart of the measured $\sim$60$\times$ probe advantage—a finite-sample realization at k=10 of the $\Theta(1/p)$ lower bound (the asymptotic factor at prevalence $p{\approx}1.17\%$ is $\sim$85$\times$): with k=0 slice labels the harm is unidentifiable (Prop. 1); a stratified probe closes the gap at the optimal label rate; and an unstratified audit provably cannot, needing $\Theta(1/p)$ times more labels. The same rarity that hides the harm from aggregates makes the *stratified* probe not just cheap but the necessary instrument (proof in Appendix).

Observability remark. Equivalently, θ's Fisher information in retained data is zero (the estimation-theoretic content of identically-distributed survivors); a size-k probe restores it to $k/(\theta(1-\theta))$ and uniform sampling dilutes that by p, the estimation complement of the Chernoff statement above.

This is where destruction bites in a way selective labels does not, and it is provable. Let M be the metadata surviving a discard.

Proposition 3 (Metadata-opaqueness makes stratification unavoidable)**.** *Let candidates be drawn i.i.d. from the population, with slice-membership indicators conditionally independent given the metadata M (each candidate's membership is fixed by its destroyed content). If S is* metadata-opaque, *$\Pr[S \mid M] = \Pr[S] = p$ a.s., then any audit that adaptively and randomly selects which examples to label as a function of the retained data, M, and the labels collected so far obtains an S-member with conditional probability at most p on every draw; hence collecting the k slice labels detection requires (Thm. 2) costs $\Omega(k/p)$ labels in expectation—no metadata proxy can reduce it, and the stratified $\Theta(k)$ rate is unattainable without an external reference that accesses S directly. More generally, if S is only approximately opaque,* $\Pr[S \mid M] \leq c\,p$

[2] Because the integer flag-threshold $\lceil \tau k \rceil / k$ oscillates around τ, h^* varies non-monotonically at small k (e.g. $0.31, 0.36, 0.37, 0.33$ at k=5, 10, 15, 20 for τ=0.35, β=0.5)—a larger probe can slightly loosen the cap, so pick k by direct computation, not by blindly increasing it.

a.s. for some $c \geq 1$, the cost degrades gracefully to $\Omega(k/(cp))$; and if S is metadata-predictable*, $\Pr[S \mid M{=}m] \geq q > p$ on an M-region of non-trivial mass, the penalty falls to $\Theta(k/q)$.*

This turns the metadata condition into a dichotomy the selective-labels setting cannot express (there the covariates X survive, so conditioning is always available): a metadata-*predictable* slice (snow, from orbit and season) is auditable at rate $\Theta(k/q)$ (why the satellite case is the conservative one), while a metadata-*opaque* slice (identity text in destroyed content) admits no proxy shortcut and forces the external reference. We measure the lift directly on the surviving metadata: conditioning the snow audit on per-patch elevation (which survives content destruction), a median-elevation$\geq$2500 m restriction raises snow prevalence from p=5.0% to q=40% in the CloudSEN12+ pool—an 8$\times$ lift, so a metadata-restricted audit collects slice labels $\sim$8$\times$ faster than uniform sampling, the $\Theta(k/q)$ branch empirically instantiated (here 5.0% is the pool prevalence, snow being deliberately over-represented in the expert pool, distinct from the 1.17% natural scene prevalence; the 8$\times$ is a within-pool ratio). The identity-text slice has no such surviving signal, so it stays at $\Theta(k/p)$. Covariate-assisted partial identification [16], [17] characterizes how surviving covariates shrink the identification *region* for θ; our contribution is the complementary *operational* quantity—the audit *label complexity*—which the metadata condition governs as a $\Theta(k)$-vs-$\Omega(k/p)$ separation, an algorithmic cost not addressed by that identification-region literature. It is the auditing content irreversibility adds *beyond* both missing labels and covariate-assisted bounds.

VIII. Limitations

The Sentinel-2 satellite snow slice is small (n=47, wide CI [66, 89]); the CloudSEN12+ rerun (n=64, five seeds), the Landsat second sensor, and the KappaSet corroboration (§IV) firm the effect, each within the caveats stated there (Landsat: 3 independent scenes; KappaSet: largest product 31%). Bare-soil (n=139) additionally replicates the targeted slice-destruction on a *different* slice (and that arm sits just under the 0.80 certification bar: it evidences the harm mechanism, not a clean certified pass), so the better-powered *certified* headline is the ingestion/curation n=192 case. Snow is intrinsically rare in CloudSEN12+ (the full expert-labeled release adds only $\sim$18 snow patches over the base set), which is why the abundant snow comes from the Landsat snow biome rather than from more Sentinel-2. The Landsat certifier is cloud-dominated, so its "certification" reflects the over-discarder being rewarded rather than a calibrated absolute bar; the tightly-controlled Landsat claim is the hidden-harm gap (1.00 vs. 0.005), not the bar.

a) Measured metadata-opaqueness of the flagship.: We asserted the moderation slice is metadata-opaque; we now measure it on the TF-IDF/logistic gatekeeper (the linear-model arm, not distilbert; membership predictor on *only* surviving non-content metadata; *muslim* slice, p=1.23%, three seeds). Attack-independent signal is weak—comment length gives AUC 0.68 (top-decile lift 2.2$\times$), the honest gatekeeper's retained score 0.74 (2.5$\times$)—so a log discarding scores sits near the opaque $\Omega(k/p)$ regime (these lifts are a *lower* bound on Prop. 3's pointwise c, which the sup over the metadata support can exceed). But the *poisoned* gatekeeper's own retained score is strongly predictive (AUC 0.95, top-decile lift 8.2$\times$): the backdoor is itself a metadata side-channel. Opaqueness is thus a property of *what the log retains*—where scores survive, slice members are findable $\sim$8$\times$ faster and the audit's search cost falls to the $\Theta(k/q)$ branch (numerically near the satellite elevation lift; a second instantiation of Prop. 3, not a violation). This shifts only label-complexity, not identifiability of θ (Prop. 1): a side-channel locates members to probe but recovers no ground truth on the destroyed discards. The strict bound is tight for logs that also discard length and scores, which we report as the conservative flagship.

Content-ingestion irreversibility is a deployment property (it holds for pre-persistence/on-device/data-curation filters, not logged post-hoc moderation). The empirical attacks are standard poisoning: the contribution is the framing, setting, identifiability, and defenses. The footprint heuristic is approximate. The label-free defense needs a majority-diverse panel; the discovery defense an enumerable protected set. Three adaptive attacks bound the defenses further: an attacker who infers τ from repeated audit feedback can sit just below it (mitigated by pairing τ with an audit cadence, as $(1-\beta)^r \to 0$); a supply-chain adversary who poisons the reference *panel* itself defeats the cross-detector (beyond the shared-blind-spot case); and harm split thinly across many latent sub-slices evades a per-slice probe, though the model-diff discovery still surfaces the aggregate removed set.

The probe defense also carries a *privacy* cost we do not resolve: drawing and labeling protected-slice examples is exactly the protected-attribute processing that data-minimization regimes restrict (the same access barrier the audit literature we cite raises [33]), so an operator's ability to run the external reference is itself constrained by privacy law, a genuine tension between auditability and data minimization.

Catastrophic harm leans adversarial; realistic mechanisms give modest (3–5$\times$) certified harm, and a deployed toxicity model showed only mild natural disparity ($\leq$1.8$\times$). The multi-generation §VI defenses (CUSUM, the adversarial game) are proof-of-mechanism, analyzed and simulated over the ratchet drift; the compounding itself is now shown in a four-generation closed-loop run (§VI), but that run is small and the CUSUM/game monitors are not yet validated on it end-to-end. The moderation dose-response, spectrum, and systemic-bias characterizations are single-seed (the satellite dose sweep a CNN, the moderation ones TF-IDF), the label-QA-evasion result specific to the reference-free outlier detector we model. Routing is a recoverable control, but it differs from the irreversible domains in task, not only in reversibility, so it is not a single-variable toggle; the tightly controlled claim is the within-domain one—harm is detectable by a retained-data audit iff the discarded data survives.

Our certification criteria are author-set and *domain-*

appropriate rather than a single uniform bar: an absolute aggregate-accuracy threshold for the onboard satellite (mirroring CloudScout's unstratified overall-accuracy validation), and a within-tolerance-of-clean criterion elsewhere (accuracy and FPR for moderation, accuracy and premium-recall for routing); each is the standard aggregate check in its domain. We note the satellite's absolute bar is the more permissive choice (a relative bar calibrated to the clean 0.825 would sit above the poison's 0.808), which is why we treat the metadata-opaque moderation case, not satellite, as the flagship. Finally, we address audit-time detection, not *training-time* robustness: interventions such as worst-group DRO or constrained ERM are orthogonal to our unidentifiability claim (which concerns what retained data can reveal) and are left to future work.

IX. Future Work

The "bound before the fact" remedy yields a concrete certification standard for irreversible gatekeepers: periodically downlink a small unfiltered random sample as the external reference. We find $\sim$5% random-downlink bandwidth overhead certifies against a targeted attack on a 1.17%-prevalence slice (97% detection / 7.5% false-alarm)—a small fraction of the bandwidth the cloud filter saves (cloudy scenes are typically the majority of captures), so the audit trims the filter's savings rather than defeating its purpose; the minimax stealth-ceiling bound we prove (Thm. 1) extends naturally to unknown-slice and multi-probe adaptive settings, an open direction; and the discovery defense should extend to open-world latent slices via representation-space or cross-model disagreement mining. The axis generalizes beyond triage (medical triage, edge-device filtering, log retention, data curation), motivating a taxonomy of deployed irreversible gatekeepers scored by exposure.

X. Ethics and Responsible Disclosure

This is an attack paper, and we treat it as dual-use. Four considerations guided us. **(1) Defense is the contribution.** The threat we describe is one defenders currently cannot detect at all—the discarded data is gone, so no existing audit applies. We publish the threat *together with* its remedy (the stratified probe, discovery scan, and certification-by-random-downlink), which is cheap in labels and deployable today where a lawful clean-reference channel exists (the data-minimization tension of Sec. VIII is the binding constraint in some regimes); on balance this arms defenders more than attackers. **(2) No real system was harmed.** All experiments run on public benchmarks (CloudSEN12, Civil Comments, RouteLLM) with synthetic poisoning; we attack a re-implementation of the published CloudScout architecture, never a live flight or production system, and compromise no user data. **(3) The attack assumes training-time access** (a strong prerequisite we state plainly), and our deeper claim is that even *non-adversarial* label bias can produce certified harm, which is a warning to system builders, not a recipe. **(4) The identity slice is a scientific instance, not a target.** A neutral-term control (the topic "water", matched $\sim$1% prevalence) is suppressed identically under the same certified attack ($0.01 \rightarrow 0.78$), confirming the mechanism is *identity-agnostic*—it destroys any rare content-defined slice; we foreground an identity term only because that is where the harm is societally consequential. We recommend operators of irreversible gatekeepers adopt the random-reference certification of Sec. IX before deployment.

XI. Conclusion

Irreversibility is a security property. When an AI gatekeeper destroys the data it rejects, targeted harm on a rare slice becomes not merely hidden but unidentifiable: certified by every standard aggregate check, invisible to every aggregate metric, and beyond the reach of every defense that assumes the data still exists. The remedy is to stop auditing the survivors and inject a small external reference. The harm cannot be measured after the fact; it must be bounded before the fact.

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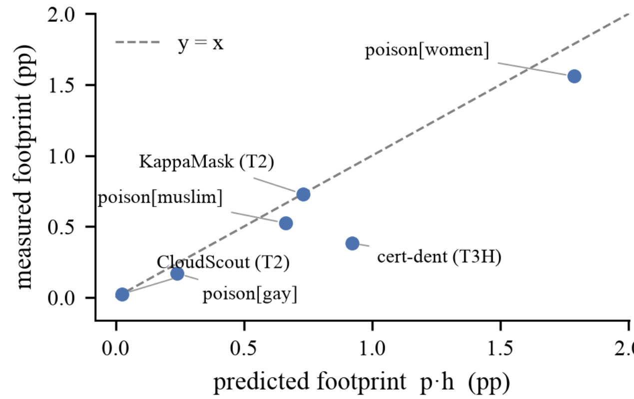

Fig. 7. Predicted footprint $p \cdot h$ vs. measured, tracking $y{=}x$; illustrates the footprint heuristic. The two satellite points are definitional (footprint was *defined* as $p \cdot h$ there); the moderation and representative-certifier points are the non-trivial fits.

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Artifact. All experiment scripts, result files, figures, and a one-command reproducibility check (`make verify`) are provided with the submission; a data-free tier reproduces the theory and defense results exactly. Every headline number in the text is a golden value read from the artifact's `results/*.json`, and the figures render from the same files. Repository link anonymized for review.

Appendix A
Footprint-Heuristic Illustration

Figure 7 plots the predicted footprint $p \cdot h$ against the measured aggregate perturbation for every case in the paper (discussion in Sec. III).

Appendix B
Proofs

Proposition 1. Write $\theta = P(\text{DISCARD} \mid \text{should-keep}, S) = D_S/(R_S + D_S)$, where R_S is the (observed) retained should-keep mass in S and D_S the (unobserved) discarded should-keep mass in S. The total discarded mass is q, so $D_S \in [0, q]$; nothing else about the discarded slice is observed. At $D_S = 0$, $\theta = 0$ (lower bound attained: the data are consistent with no false discards). The only unobserved quantity is D_S; R_S is fixed by the retained data. Since $\theta = D_S/(R_S + D_S)$ is increasing in D_S over $D_S \in [0, q]$, the identified set is obtained by ranging D_S over its bounds: $\theta \in [\,0,\ q/(R_S + q)\,]$. Writing $R_S = a(1-q)$, where a is the *observed* fraction of retained inputs that lie in S and are should-keep (so R_S is the retained should-keep-S mass), the upper endpoint is $q/(a(1-q)+q)$. Both endpoints are attained and no statistic of the retained data distinguishes points in the interval, so θ is only partially identified and $\theta > 0$ cannot be certified. $\square$

Theorem 1. A size-k probe drawn from S observes X flagged false-discards; under true slice-harm h, $X \sim \text{Bin}(k, h)$, and the defender flags iff the observed rate is $\geq \tau$, i.e. $X \geq \lceil \tau k \rceil =: m$. Detection probability is

$g(h) = \Pr[\mathrm{Bin}(k,h) \geq m]$. For fixed $k, m \geq 1$, g is strictly increasing on $[0,1]$ (a $\mathrm{Bin}(k,h)$ stochastically dominates $\mathrm{Bin}(k,h')$ for $h > h'$), with $g(0) = 0$ and $g(1) = 1$; hence for any $\beta \in (0,1)$ there is a unique h^* with $g(h^*) = \beta$, and $\{h : g(h) \leq \beta\} = [0, h^*]$. The attacker's maximum harm subject to detection $\leq \beta$ is therefore $h^*(k, \tau, \beta) = \max\{h : g(h) \leq \beta\}$, computable by bisection since g is monotone. As $k \to \infty$ the empirical rate $X/k \to h$ almost surely, so $g(h) \to \mathbf{1}[h \geq \tau]$; keeping $g(h) \leq \beta < 1$ then forces $h \leq \tau$, giving $h^* \to \tau$. □

Theorem 2. Each labeled slice example is $\mathrm{Bern}(b)$ under a clean slice and $\mathrm{Bern}(h)$ under an attacked slice; an audit sees k i.i.d. such labels. By Chernoff's theorem for symmetric hypothesis testing [18], [19], the minimum total error of any test of $\mathrm{Bern}(b)^{\otimes k}$ vs $\mathrm{Bern}(h)^{\otimes k}$ is $e^{-kC(b,h)(1+o(1))}$, where $C(b,h) = -\log \min_{s\in[0,1]} \left(b^s h^{1-s} + (1-b)^s(1-h)^{1-s}\right)$ is the Chernoff information; no test decays faster. To reach error ε we thus need $k \geq \log(1/\varepsilon)/C(b,h)\,(1+o(1))$, i.e. $k = \Theta(\log(1/\varepsilon)/C)$ labeled slice examples (necessity up to lower-order terms). For the threshold probe "flag if empirical rate $\widehat{r} \geq \tau$", Cramér's theorem gives the false-positive tail $\Pr[\widehat{r} \geq \tau \mid b] = e^{-kD(\tau\|b)(1+o(1))}$ and the false-negative tail $\Pr[\widehat{r} < \tau \mid h] = e^{-kD(\tau\|h)(1+o(1))}$, so the probe's total-error exponent is $\min\{D(\tau\|b), D(\tau\|h)\}$. On $\tau \in (b,h)$, $D(\tau\|b)$ is increasing and $D(\tau\|h)$ decreasing, so this minimum is maximized where the two are equal, at τ^*; at that point the common value is exactly $C(b,h)$. To see this, let s^* minimize $C(b,h) = -\log \sum_x b(x)^s h(x)^{1-s}$ and let $P_{s^*}(x) \propto b(x)^{s^*} h(x)^{1-s^*}$ be the corresponding tilted (geometric-mean) law—a $\mathrm{Bern}(\tau^*)$ with $\tau^* = b^{s^*}h^{1-s^*}/Z$, $Z = b^{s^*}h^{1-s^*} + (1-b)^{s^*}(1-h)^{1-s^*}$. The first-order condition $\partial_s \log Z = 0$ is precisely $D(P_{s^*}\|b) = D(P_{s^*}\|h)$, and each equals $-\log Z = C(b,h)$; since $P_{s^*} = \mathrm{Bern}(\tau^*)$ this reads $D(\tau^*\|b) = D(\tau^*\|h) = C(b,h)$ [19], Thm. 11.9. As τ^* is the equal-KL crossing, the threshold probe there attains exponent C and is rate-optimal, matching the lower bound. For the last claim, an audit that samples the population *uniformly* draws a slice member (prevalence p) with probability p per draw, so the expected draws to collect k slice labels is k/p: any such unstratified audit needs $\Theta(k/p)$ labels, a factor $1/p$ more. (This necessity is stated for uniform population sampling; a procedure able to identify slice members from surviving metadata could evade it—but by the scope condition, for a metadata-opaque slice no such shortcut exists.) □

Proposition 3. Index the labels the audit collects $t = 1, 2, \ldots$, and let $\mathcal{F}_t$ be the σ-field of everything the audit has seen through label t: the retained data, M, and the membership indicators $Y_1, \ldots, Y_t$ of the candidates it already labeled. An adaptive rule chooses the $(t+1)$-th candidate as an $\mathcal{F}_t$-measurable function. Every component of $\mathcal{F}_t$ is uninformative about Y_{t+1} beyond M: past labels by conditional independence of memberships given M, and the retained data (a function of *other* i.i.d. candidates) by the independence of distinct draws given M. Hence, with opaqueness, $\Pr[Y_{t+1}{=}1 \mid \mathcal{F}_t] = \Pr[Y_{t+1}{=}1 \mid M] = p$. Thus $H_t = \sum_{u\leq t} Y_u - pt$ is an $\mathcal{F}_t$-martingale. Let $\sigma_k = \inf\{t : \sum_{u\leq t} Y_u = k\}$ be the first time k slice labels are collected. Since each conditional hit probability is $p > 0$, σ_k is stochastically dominated by a negative-binomial(k,p) and hence $\mathbb{E}[\sigma_k] < \infty$; the increments $Y_{t+1} - p \in [-p, 1-p]$ are bounded. Bounded increments together with $\mathbb{E}[\sigma_k] < \infty$ satisfy the Wald/optional-stopping criterion, giving $\mathbb{E}[H_{\sigma_k}] = 0$, i.e. $k = p\,\mathbb{E}[\sigma_k]$, hence $\mathbb{E}[\sigma_k] = k/p$: any adaptive metadata-measurable audit needs $\Omega(k/p)$ labels, and the stratified $\Theta(k)$ rate requires drawing from S directly (an external reference). Under approximate opaqueness $\Pr[Y_{t+1}{=}1 \mid \mathcal{F}_t] \leq cp$, the process $\sum_{u\leq t} Y_u - cpt$ is a *supermartingale*; σ_k has finite mean (fresh population candidates are in S with marginal probability $p > 0$ independent of the history, so σ_k is again dominated by a negative-binomial(k,p)), and the optional-stopping inequality for supermartingales (bounded increments, $\mathbb{E}[\sigma_k] < \infty$) then gives $\mathbb{E}[\sum_{u\leq\sigma_k} Y_u] \leq cp\,\mathbb{E}[\sigma_k]$, i.e. $k \leq cp\,\mathbb{E}[\sigma_k]$, so $\mathbb{E}[\sigma_k] \geq k/(cp)$. Under metadata-predictability, restricting selection to a region R with $\Pr[S \mid M{\in}R] \geq q$ (and $\Pr[M \in R]$ bounded below) raises the per-label hit probability to $\geq q$, giving $\mathbb{E}[\sigma_k] = \Theta(k/q)$. □